

PDE3 May 2022 Exam Question 2

(b) Elliptical coordinates on the plane (x, y) are:

$$\left. \begin{aligned} x &= c \cosh \mu \cos \nu \\ y &= c \sinh \mu \sin \nu \end{aligned} \right\}$$

Figure QA2 shows the grid for such a coordinate system. Lines of constant μ are ellipses with foci at $(-c, 0)$ and $(c, 0)$, while lines of constant ν are hyperbolas with the same foci. The Laplace equation transformed into this coordinate system is:

$$0 = \frac{1}{c^2 (\sinh^2 \mu + \sin^2 \nu)} \left[\frac{\partial^2 u}{\partial \mu^2} + \frac{\partial^2 u}{\partial \nu^2} \right]. \quad (\text{Q2-4})$$

Using appropriate finite difference approximations, discretise Equation Q2-4 and hence determine the coefficients associated with the five points in the stencil u_N , u_E , u_S , u_W and u_O . (6)

(c) In the computational domain you are solving the equations on a uniform grid with mesh spacing $\Delta \nu$ in the i direction and $\Delta \mu$ in the j direction. Sketch the computational domain and briefly explain what type of boundary condition is needed on the West, $\nu = 0$, and East, $\nu = 2\pi$ boundaries and how you would implement it. (4)

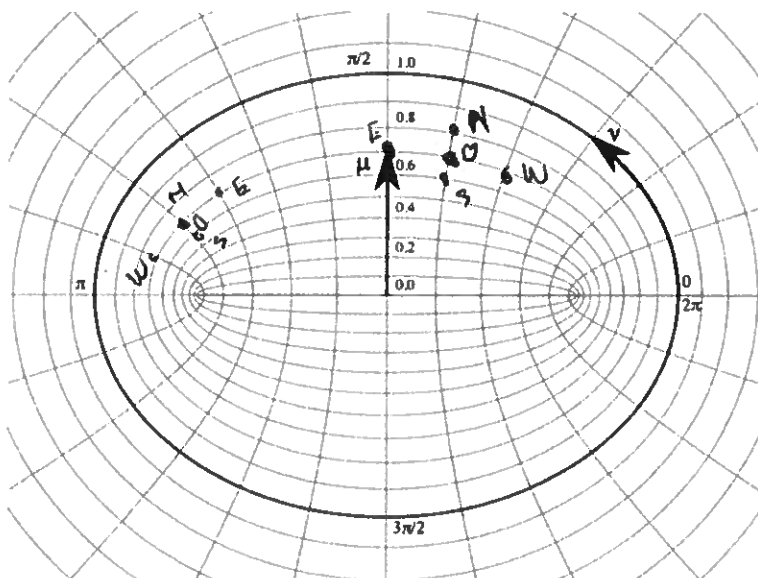


Figure QA2: Elliptical (μ, ν) coordinates on the plane.

May 2022 Exam Question 2

b) Use central difference schemes

$$\frac{\partial^2 u}{\partial \mu^2} = \frac{u_N - 2u_0 + u_S}{\Delta \mu^2}$$

$$\frac{\partial^2 u}{\partial v^2} = \frac{u_E - 2u_0 + u_W}{\Delta v^2}$$

→ Jacobi ~~scheme~~ scheme:

$$\alpha_N u_N + \alpha_S u_S + \alpha_W u_W + \alpha_E u_E + \alpha_0 u_0 = 0$$

FD scheme for the partial μ derivative:

$$\frac{1}{c^2 (\sinh \mu + \sin^2 v)} \frac{\partial^2 u}{\partial \mu^2} = \left[\frac{1}{c^2 (\sinh \mu_N + \sin^2 v_N)} \frac{u_N}{\Delta \mu^2} \right] \Rightarrow \alpha_N$$

$$- \frac{2}{c^2 (\sinh \mu_0 + \sin^2 v_0)} \frac{u_0}{\Delta \mu^2}$$

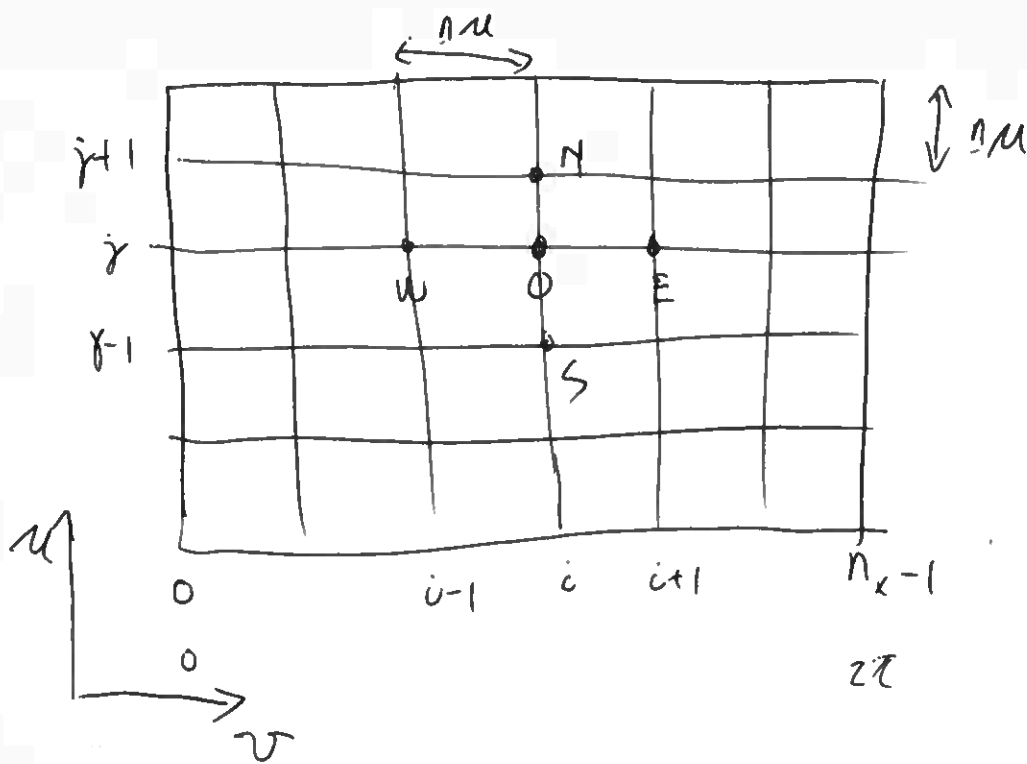
$$+ \left[\frac{1}{c^2 (\sinh \mu_S + \sin^2 v_S)} \frac{u_S}{\Delta \mu^2} \right] \Rightarrow \alpha_S$$

$$\frac{1}{c^2 (\sinh \mu + \sin^2 v)} \frac{\partial^2 u}{\partial v^2} = \left[\frac{1}{c^2 (\sinh \mu_E + \sin^2 v_E)} \frac{u_E}{\Delta v^2} \right] \Rightarrow \alpha_E$$

$$- \frac{2}{c^2 (\sinh \mu_0 + \sin^2 v_0)} \frac{u_0}{\Delta v^2}$$

$$+ \left[\frac{1}{c^2 (\sinh \mu_W + \sin^2 v_W)} \frac{u_W}{\Delta v^2} \right] \Rightarrow \alpha_W$$

$$\alpha_0 = - \frac{2}{c^2 (\sinh \mu_0 + \sin^2 v_0)} \frac{1}{\Delta \mu^2} - \frac{2}{c^2 (\sinh \mu_0 + \sin^2 v_0)} \frac{1}{\Delta v^2}$$



For $v=0$ the west BC $u_{i=0,W} = u_{i=n_x-1}$

For $v=2\pi$ the east BC $u_{i=n_x-1,E} = u_{i=0}$

PDE3 December 2017 Exam Question 3

Question 3. The governing equation for the one dimensional, transient, heat transfer equation is:

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\kappa} \frac{\partial T}{\partial t},$$

where T is the temperature in Kelvin, and κ is the thermal diffusivity of the material (in m^2s^{-1}).

- (a) Using appropriate finite difference approximations for the derivatives show that the forward time centred space (FTCS) scheme for the solution of the governing equation is,

$$T_i^{n+1} = T_i^n + \frac{\kappa \Delta t}{\Delta x^2} (T_{i+1}^n - 2T_i^n + T_{i-1}^n),$$

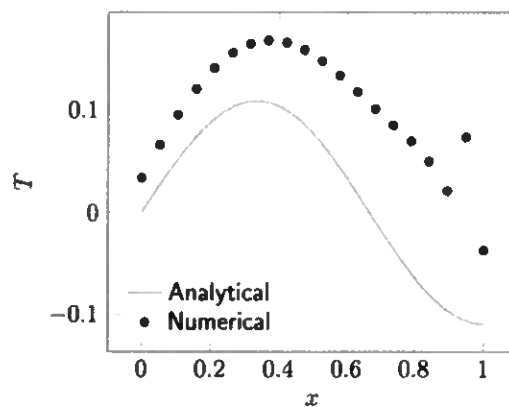
where $T_i^n = T(x_i, t_0 + n\Delta t)$ and, state the formal order of accuracy of the scheme in time and space. (6)

- (b) Using von Neumann stability analysis shows that the FTCS scheme is stable if and only if

$$\boxed{\Delta t \leq \frac{\Delta x^2}{2\kappa}} \Rightarrow \Delta t = \nu \frac{\Delta x^2}{2\kappa} \quad \text{L} \rightarrow \text{Courant number (10)}$$

- (c) To avoid too restrictive a time step, it has been suggested that the Crank-Nicolson scheme be employed which is unconditionally stable. Figure Q3-1 shows the results obtained using a Courant number of 50. Why is the solution different from the analytical solution and how might the solution be improved? (4)

next week.



stability $\neq$ accuracy
 $\rightarrow$ increase time resolution
 decrease Δt
 $\rightarrow$ increase spatial resolution
 decrease Δx .

Figure Q3-1: Comparison of analytical and numerical solutions for a heat transfer problem solved with the Crank-Nicolson scheme with $\nu = 50$.

(d) Consider the steady-state heat conduction equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

which has been solved numerically for a certain set of boundary conditions using the following Successive Over-Relaxation (SOR) scheme

$$u_O^{n+1} = (1 - \omega)u_O^n + \frac{\omega}{2(1 + \beta^2)} (u_E^n + u_W^{n+1} + \beta^2(u_N^n + u_S^{n+1})) \quad (6)$$

The following integrated quantity of the solution u

$$U = \int_{\Omega} u^2 dy dx$$

has been evaluated at four grids with different sizes. Table Q1 shows the calculated values of U for each grid from finer to coarser, the grid sizes and the spatial step of each grid. Calculate the order of convergence of the numerical scheme. What is the theoretical order of convergence expected by this scheme? Do the theoretical and calculated values agree? Justify your answer.

(5)

	Grid size	U	$\Delta x(m)$
1)	321×161	0.161197	0.00625
2)	161×81	0.161296	0.0125
3)	81×41	0.161604	0.025
4)	41×21	0.162798	0.05

} most accurate ones

Table Q1: Grid Convergence Analysis values

(e) Following part (d) and using the values from Table Q1: Is the fourth grid in the asymptotic range? Justify your answer. Briefly mention what it means for a solution to be at the asymptotic range.

(5)

$$U(\Delta x) = \underbrace{U(\Delta x \rightarrow 0)}_{\text{Richardson extrapolation}} + C (\Delta x)^p \rightarrow \text{order of convergence.}$$